Engine Model(s): PT6A

72-30-05

COMPRESSOR ROTOR - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Maintenance of the compressor rotor is restricted to inspection and repair of the first-stage rotor.

C. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

D. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

Not Applicable

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Inspection/Check- (First-stage Compressor Rotor)

A. General

CAUTION: KEEP COMPRESSOR INLET AREA CLEAR OF ALL FOREIGN OBJECTS THAT COULD BE INGESTED BY ENGINE AND CAUSE DAMAGE DURING SUBSEQUENT OPERATION.

NOTE: The following inspection should be done whenever the air inlet screen is removed (Ref. 72-20-00). Due to limited access, the use of a mirror is recommended.

(1) The limits stated under 'Acceptable Damage Limits' and 'Acceptable Erosion Limits' are considered acceptable without repair. The actual damage must be documented and additional inspections scheduled to assess the rate of defect progression.

NOTE: If acceptable limits are exceeded, the blades may still be repaired (Ref. Approved Repairs).

(2) Continued operation of a compressor with FOD, either accepted as-is or blended is dependent on satisfactory engine performance.

NOTE: When assessing damage, it should be understood that continued operation with a large number of blades damaged or eroded at or near maximum acceptable limits

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may result in reduction of compressor efficiency and engine performance.

(3) Acceptable damage limits are based primarily on structural integrity considerations. Rotors with damage exceeding 'Acceptable Damage Limits' cannot remain in service unless damage can be removed by blending (Ref. Approved Repairs):

Table 201 Example of FOD (Ref. Fig. 203, Sheet 1)

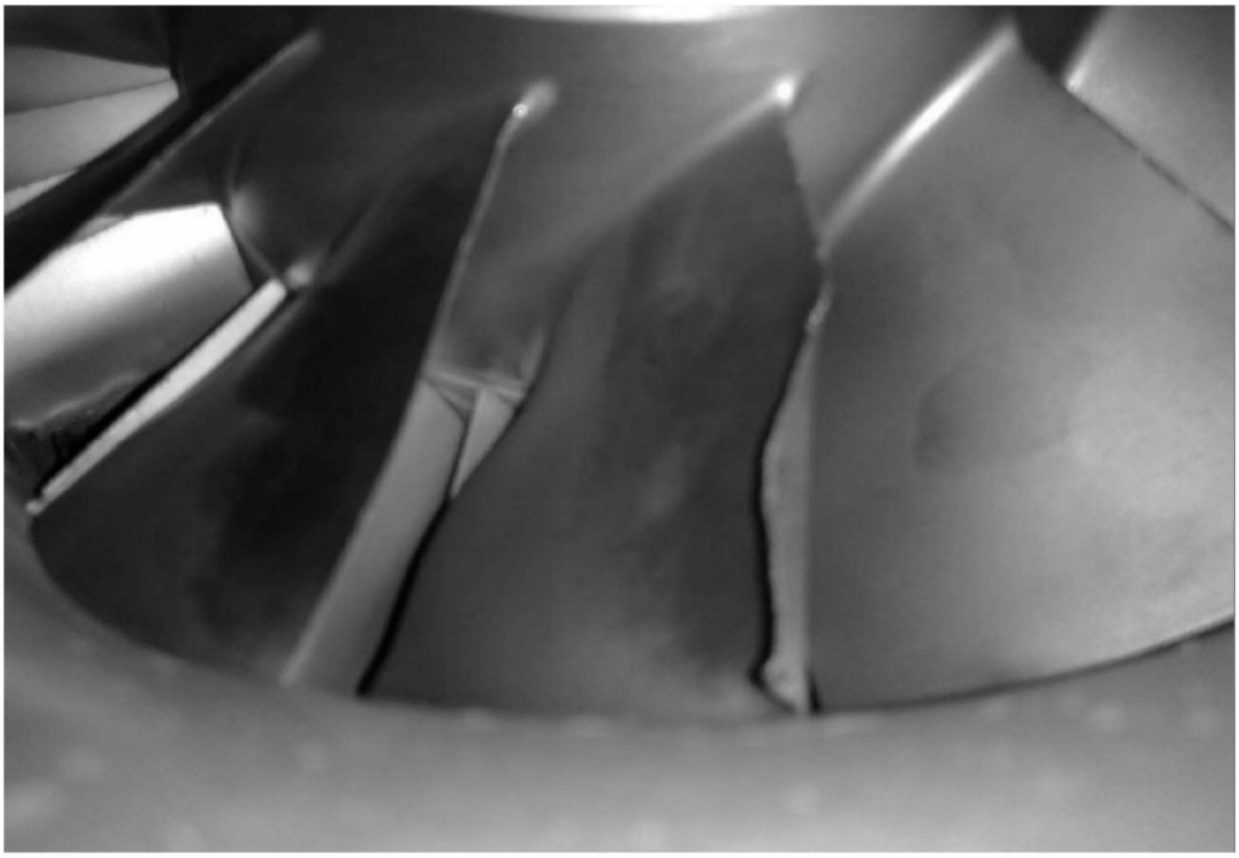
Type of Damage	Disposition
A nick, 0.020 inch deep by 0.025 inch wide, in blade root.	Engines may not continue in operation with damage in blade root. However, if the damage can be removed by performing a leading edge cutback (Ref. Approved Repairs) the engine can return to service provided performance is satisfactory.

NOTE: Remove the minimum material required to remove the damage. Document the damage (Ref. Subpara. B.).

(4) If large bent airfoil (Ref. Fig. 201) causes the engine to make unusual compressor noise or if you find the blade damage is more than the acceptable limits (Ref. Para. 5. C.), remove and send the engine to a P&WC approved overhaul facility for light overhaul inspection of large bent airfoil.

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Figure 201 Large Bent Airfoil on First-Stage Compressor Rotor



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(5) If damaged blade is within acceptable limits and is not reworked, inspect area at 100 hours. Subsequent inspections are at the discretion of the operator but must not exceed 400 hours.

(6) If damage to the blades is acceptable, and/or after blending, engine performance is within acceptable limits, inspection of the subsequent stages, (rotors, stators and impeller) is not required. Under these conditions, the engine is considered acceptable for service.

(7) If damage exceeds limits and cannot be removed by blending or engine performance is unsatisfactory after repair, the engine or gas generator must be removed from service.

B. Documenting Damage

(1) Defects/damage should be documented (i.e., engine module TSN/TSO, cycles, component description, location and dimensions, if obtainable).

(2) If the blade is repaired document the amount of material removed, (i.e. if the blade was cut back, the amount of cutback should be documented).

C. Acceptable Damage Limits (Ref. Fig. 202)

CAUTION: NICKS IN LEADING AND TRAILING EDGES BECOME INCREASINGLY CRITICAL TOWARD ROOT. DAMAGE IS NOT PERMITTED WITHIN 0.050 INCH OF BLADE ROOT FILLET RADIUS. THE ENGINE MAY REMAIN IN SERVICE PROVIDED THE DAMAGE IS COMPLETELY REMOVED BY LEADING EDGE CUTBACK.

(1) Multiple leading edge nicks 0.050 inch deep on any number of blades are acceptable.

(2) Leading or trailing edge nicks 0.100 inch deep maximum. Two per blade on any number of blades.

NOTE: A nick is considered to be a sharp surface indentation caused by impact resulting in loss of parent material.

(3) Leading or trailing edge dents 0.100 inch deep by 0.100 inch long. Two per blade on any number of blades.

(4) Leading or trailing edge tip curl, without tearing, maximum is 0.100 inch deformation on one edge, up to four blades.

(5) Leading edge nicks 0.030 to 0.050 inch deep by 0.080 inch wide, more than 0.200 inch apart, and more than 0.250 inch from blade tip, on any number of blades, are acceptable provided there are no tears and/or cracks.

(6) Trailing edge dents and nicks, maximum of four (4) per blade, 0.030 to 0.050 inch deep by 0.080 inch wide, more than 0.200 inch apart, and more than 0.250 inch from blade tip, on any number of blades, are acceptable provided there are no tears or cracks.

(7) Blade tip rub is acceptable providing there is no rolled over material and an engine performance check is acceptable.

D. Acceptable Erosion Limits

(1) Inspect first-stage compressor blades for erosion of leading edge.

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(2) Erosion at the root is acceptable providing the chord is not reduced by more than 0.250 inch and there is no feathered, (i.e., sharp) edge(s).

NOTE: The chord of an airfoil section is the shortest distance between the leading and trailing edges.

(3) Feathered edges must be removed (Ref. Approved Repairs) or the engine or gas generator module removed from service. The original airfoil chord of the repaired section must not be reduced by more than 0.250 inch and the original leading edge profile must be restored.

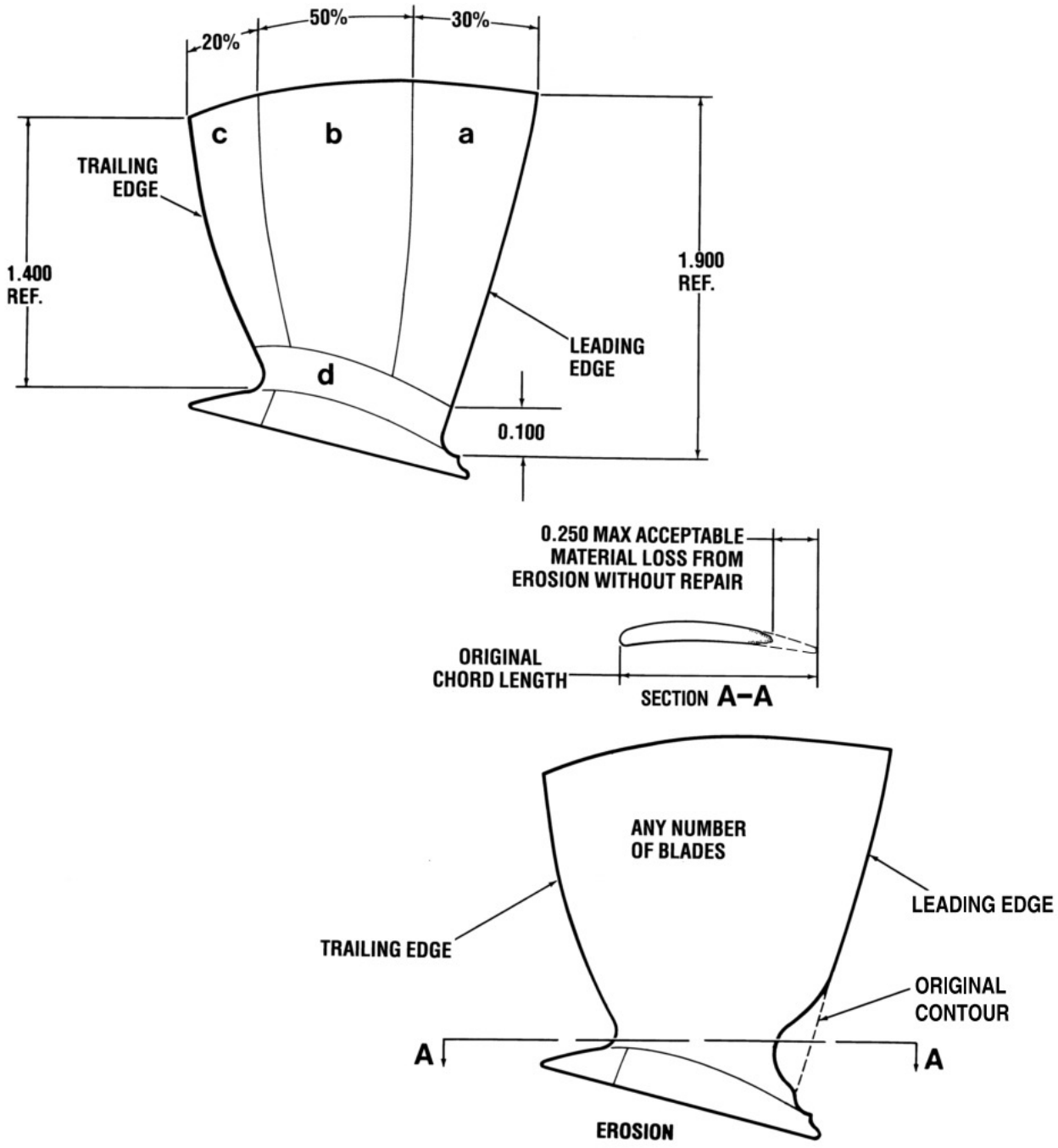
(4) Blade tip rub is acceptable, providing an engine performance check (Ref. Aircraft Maintenance Manual) is acceptable and there is no "rolled" over material at the blade tip.

E. Unacceptable Damage (Ref. Fig. 203)

(1) Tears and cracks due to deformation of leading or trailing edges and cracks on any blade surface are not acceptable. The blade must be repaired if the defect is within repairable limits (Ref. Approved Repairs), or the engine or gas generator module removed from service.

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Figure 202 Compressor First-stage Blades Inspection

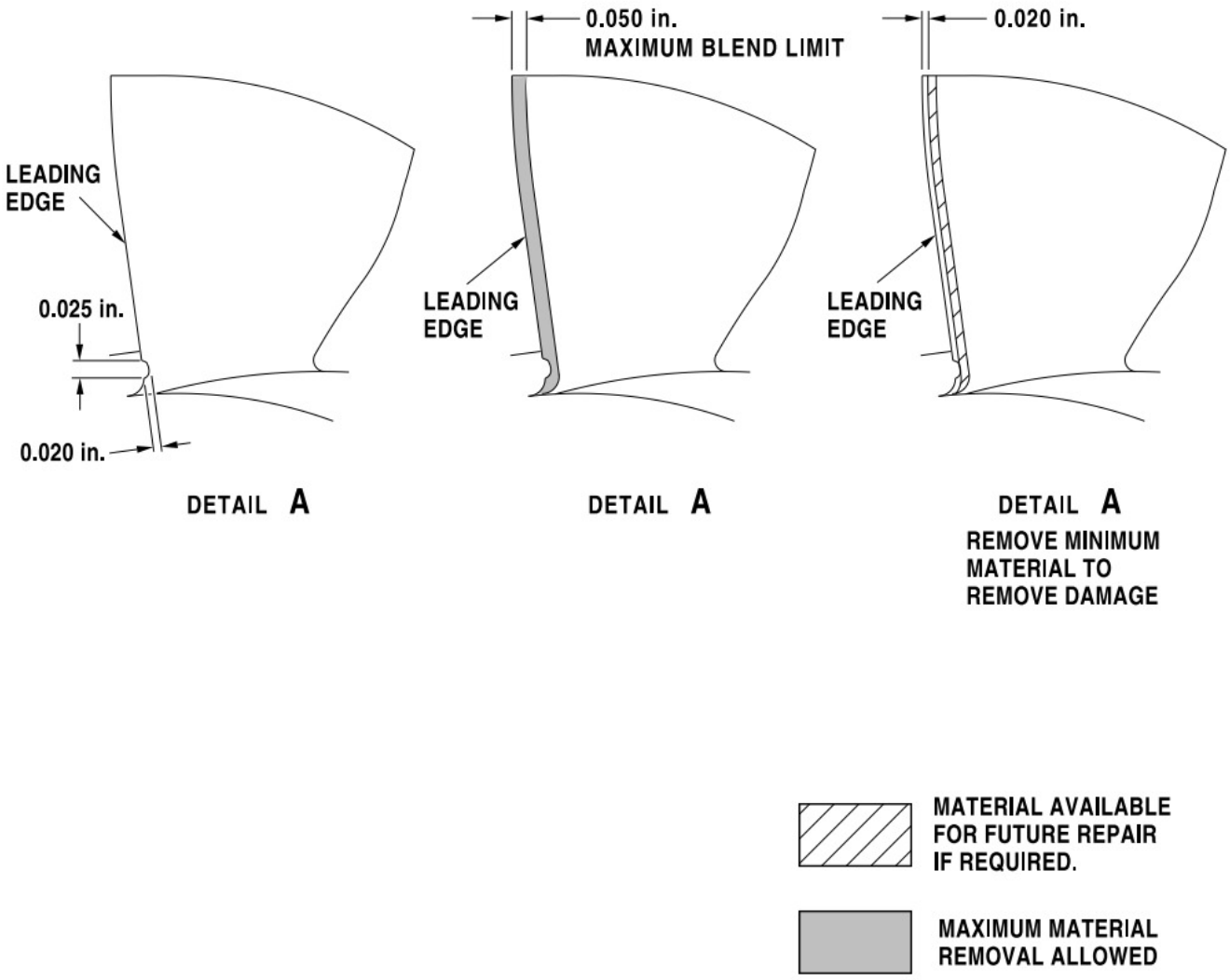


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Figure 203 Example of Damage Removal

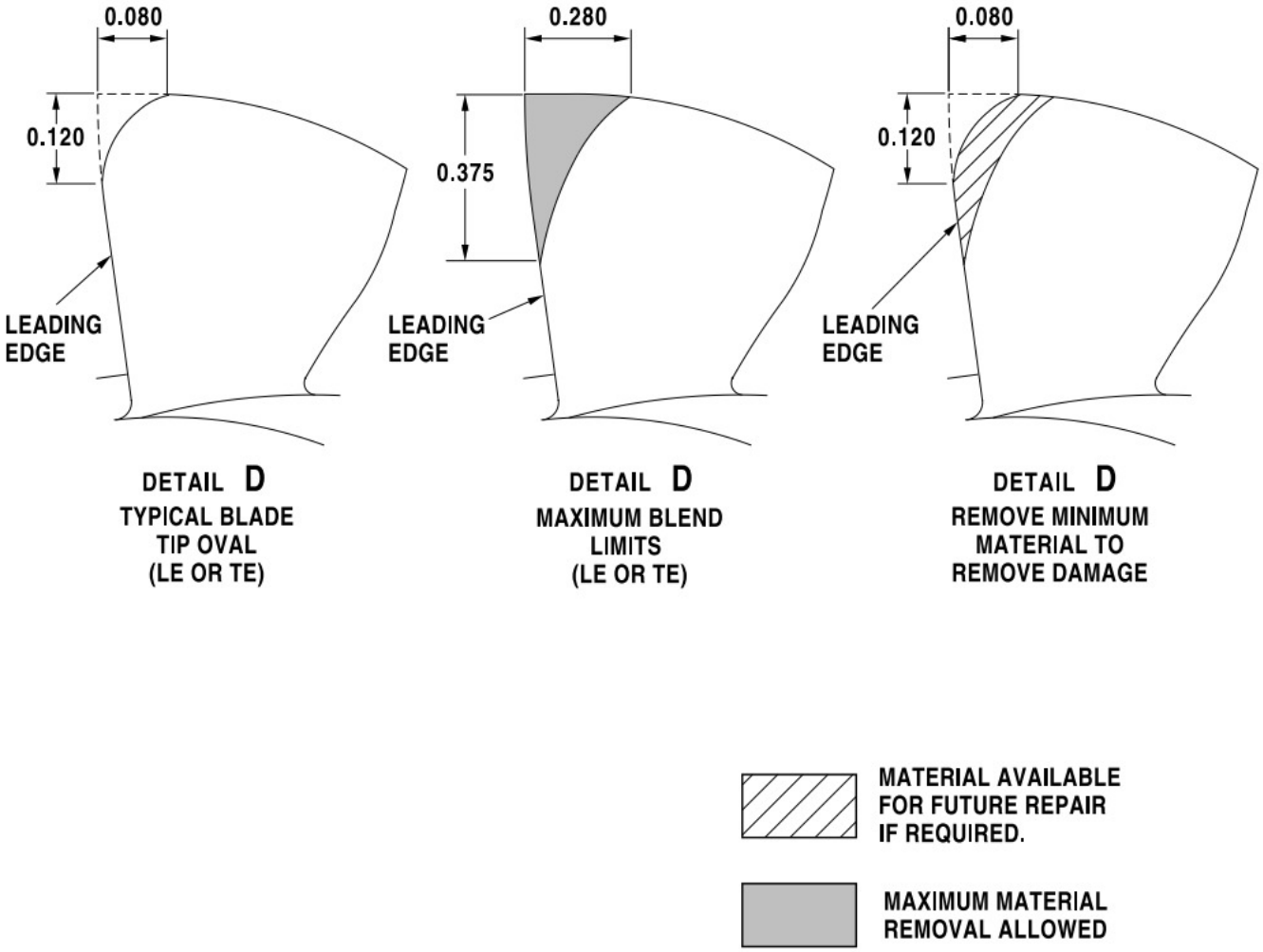
(SHEET 1 OF 3)



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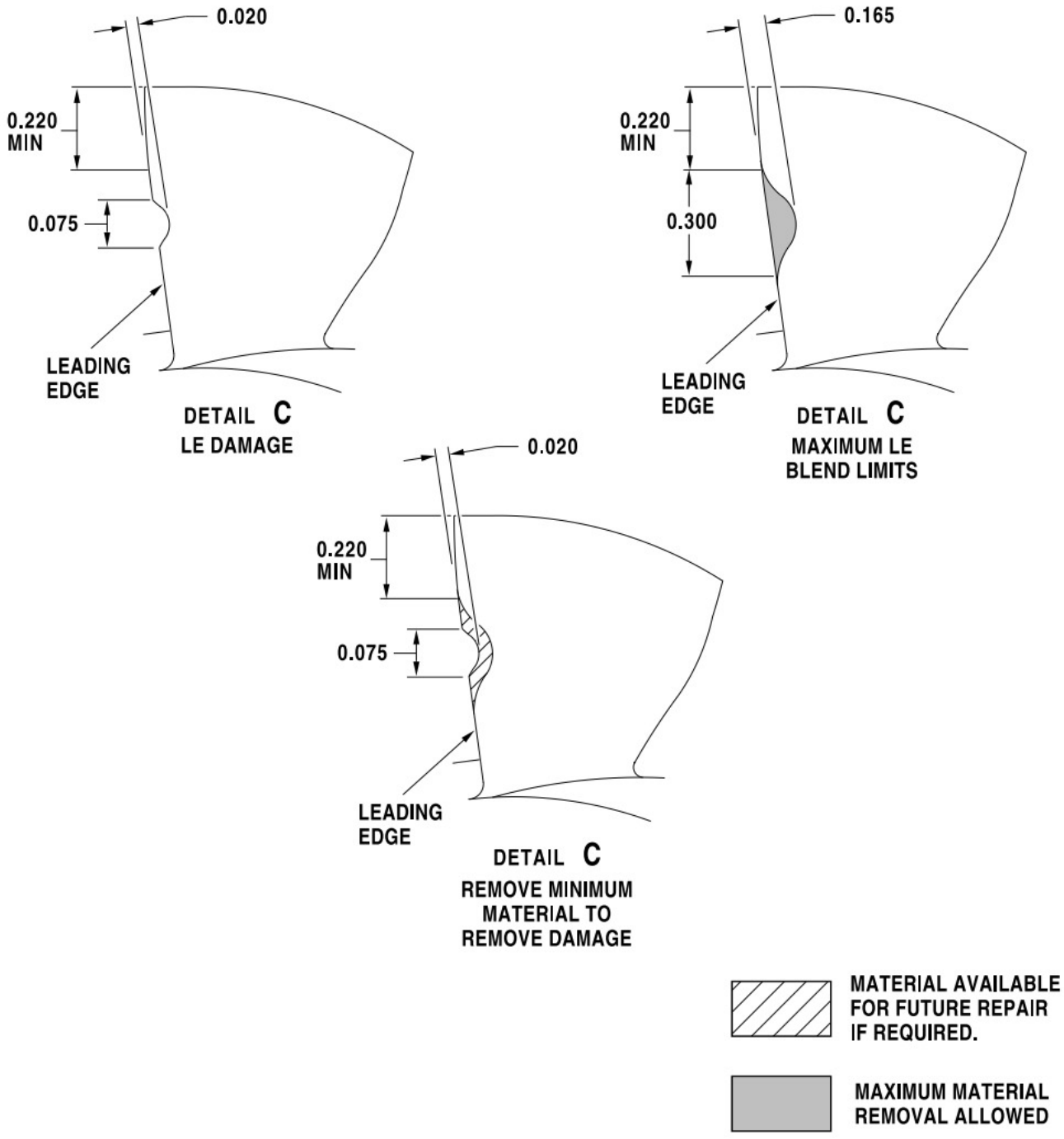
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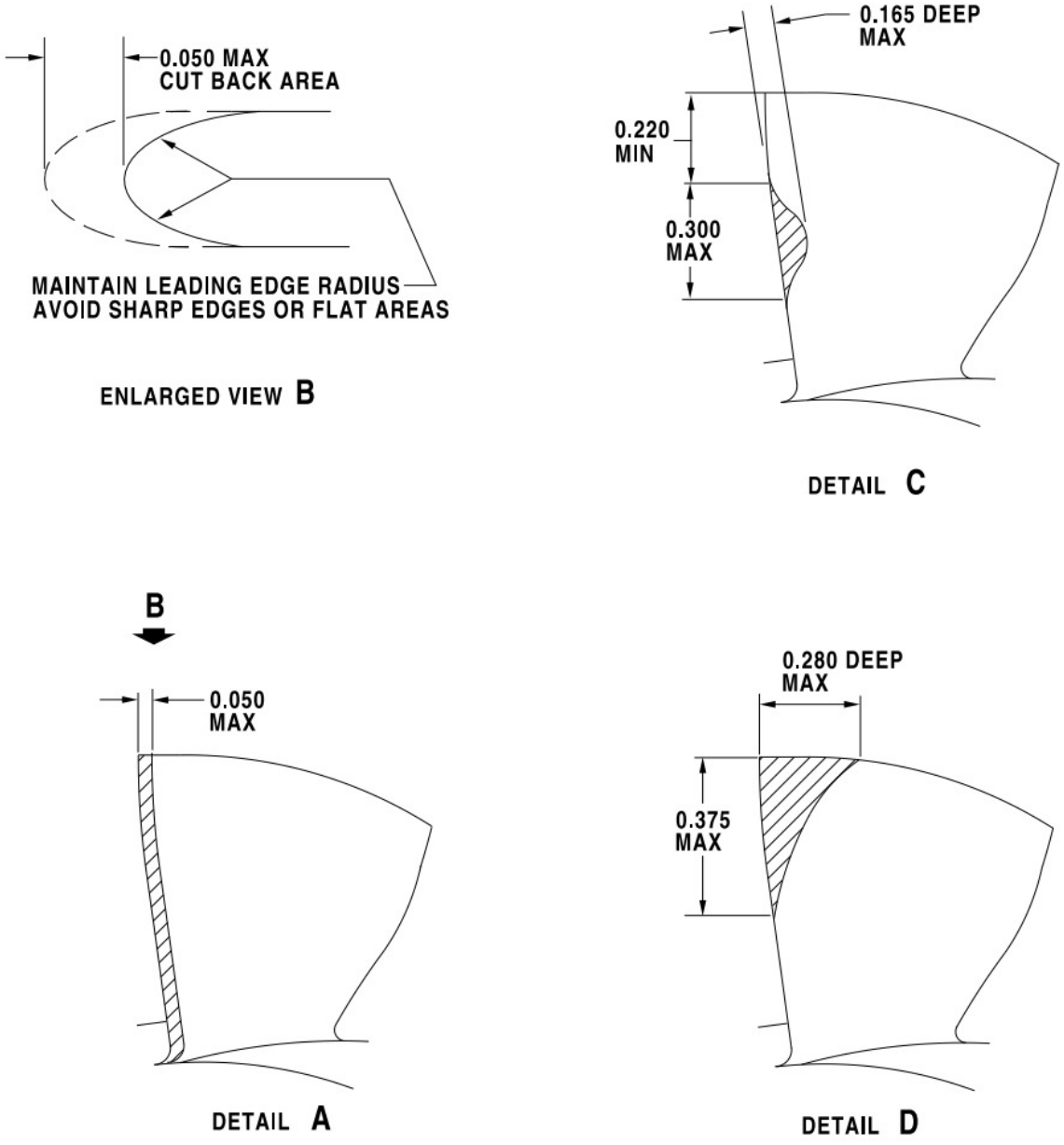
6. Approved Repairs - (First-stage Compressor Rotor)

A. Repair of Leading Edge (Ref. Fig. 204)

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Figure 204 Compressor First-stage Repair Limits

(SHEET 1 OF 2)

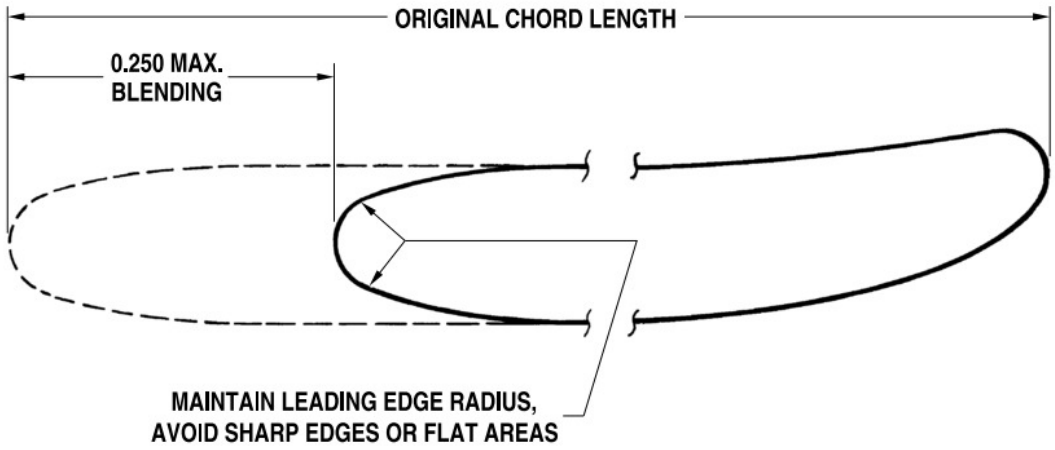


NOTE: THE MAXIMUM PERMITTED MATERIAL REMOVAL IS SHOWN. ONLY REMOVE THE MINIMUM MATERIAL NECESSARY TO BLEND OUT DAMAGE.

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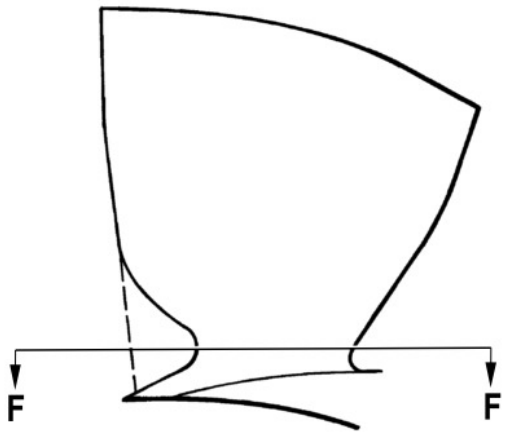
(SHEET 2 OF 2)

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ENLARGED SECTION **F–F**

NOTE: THE MAXIMUM PERMITTED MATERIAL REMOVAL IS SHOWN. ONLY REMOVE THE MINIMUM MATERIAL NECESSARY TO BLEND OUT DAMAGE.



DETAIL **E**

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CAUTION: BLENDING OF THE FIRST-STAGE COMPRESSOR ROTOR BLADES PRIMARILY AFFECTS ENGINE PERFORMANCE AT ALTITUDE. AFTER BLENDING VERIFY ENGINE PERFORMANCE ON THE GROUND.

CAUTION: KEEP THE COMPRESSOR INLET AREA CLEAR OF ALL FOREIGN MATERIAL AND OBJECTS WHICH COULD BE INGESTED BY THE ENGINE AND CAUSE SERIOUS DAMAGE DURING OPERATION.

(1) Repair the first-stage blade leading edge within the following limits using fine hand files and stones.

WARNING: AN ACCUMULATION OF TITANIUM DUST IS A FIRE HAZARD. CLEAN UP TITANIUM DUST ON COMPLETION OF REPAIR AND DO NOT ALLOW DUST TO ACCUMULATE ON CLOTHING.

(2) Suitably mask the entire first-stage compressor area to prevent contamination of the compressor with filing residue.

(3) In case of multiple leading edge defects situated closer than 0.200 inch, the entire leading edge of an unlimited number of blades may be cut back 0.025 inch maximum (Detail A). If blade has been previously cut back, reduce repair limit by previous cut back. Maintain the leading edge profile and fillet radius as close as possible to the original.

NOTE: If cutting back the leading edge by 0.025 inch does not eliminate or make defects acceptable, the blade cutbacks may be extended to 0.050 inch maximum, providing engine performance is within limits. This is acceptable to permit the engine to remain in service; however, the affected components may have to be replaced at the next overhaul.

(4) Nicks and tears may be smoothed to 0.300 inch wide maximum by 0.165 inch deep maximum, provided repaired area is 0.220 inch minimum from blade tip (Detail C).

NOTE: 1. If repairing within the above limits does not eliminate the defects or make them acceptable, individual defects may be smoothed to 0.500 inch wide maximum and 0.250 inch deep maximum, provided repair area is 0.350 inch minimum from blade tip, and engine performance is within limits. However, the affected components may have to be replaced at next overhaul.

NOTE: 2. If any blade is smoothed to the maximum limit, the blade 180 degrees opposite must be smoothed to the same shape to prevent compressor rotor imbalance.

(5) Bent or torn leading edge tips may be repaired to 0.375 inch wide maximum by 0.280 inch deep maximum (Detail D).

NOTE: If any blade is smoothed to the maximum limits, the blade 180 degrees opposite must be smoothed to the same shape to prevent compressor rotor imbalance.

(6) Smooth sharp edge at the blade leading edge root eroded area to restore as near as possible the original radius and smoothness (Detail E). After repair, make sure the original airfoil chord at the repair area was not reduced by more than 0.250 inch.

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CAUTION: KEEP COMPRESSOR AIR INLET AREA CLEAR OF ALL FOREIGN MATERIAL AND OBJECTS WHICH COULD BE INGESTED BY THE ENGINE AND CAUSE SERIOUS DAMAGE DURING ENGINE OPERATION.

(7) Using a suitable suction cleaner, thoroughly clean entire compressor inlet area. Remove masking material and repeat cleaning.

B. Repair of First-stage Blade Trailing Edge

(1) Dents and nicks up to 0.050 inch deep and wide, not less than 0.200 inch apart and not within 0.250 inch of blade tip, are acceptable (maximum 4 per blade), providing there are no tears or cracks.

(2) Sharp corners must be smoothed to an approximate radius of 0.050 inch and a maximum of 0.050 inch deep and wide. Tip corner radius must not exceed 0.100 inch

CAUTION: KEEP COMPRESSOR INLET AREA CLEAR OF ALL FOREIGN OBJECTS WHICHCOULD BE INGESTED BY ENGINE AND CAUSE DAMAGE DURING ENGINE OPERATION.

(3) Using suitable suction cleaner, thoroughly clean entire compressor inlet area. Remove all masking material and repeat cleaning. Make sure all filing residue is removed from inlet area and first-stage compressor blades.

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